

Indexed sets:-

A, B, C, D, E, F, G, ...

$A_1, A_2, A_3, A_4, \dots, A_n$

$$A_1 = \{1, 2, 3\}$$

$$A_2 = \{3, 4, 5\}$$

$$A_3 = \{5, 6, 7\} \dots A_n$$

$$A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n$$

$$\bigcap_{i=1}^n A_i$$

$$1 + 2 + 3 + \dots + 60$$

$$\sum_{n=1}^{60} n$$

$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n$$

$$\text{ex:- } A_1 = \{0, 2, 5\} \quad A_2 = \{4, 7\} \quad A_3 = \{2, 4, 11\}$$

$$A_4 = \{3, 6, 9\} \quad A_5 = \{6, 8, 10\}$$

$$A_6 = \{1, 2, 3, 4, 5\}$$

$$\bigcup_{i=1}^4 A_i = A_1 \cup A_2 \cup A_3 \cup A_4 = \{0, 2, 5, 4, 7, 11, 6, 3, 9\}$$

$$\bigcap_{i=3}^6 A_i = A_3 \cap A_4 \cap A_5 \cap A_6 = \emptyset$$